

FL Simulation Results

File: test_B2

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 4
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 50.0% (0.50)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 50.0% (0.50) of data for 15 rounds with 4 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8758
Clean + DP Protection Accuracy: 0.8762
Poisoned Accuracy: 0.8445
Poisoned + DP Protection Accuracy: 0.8445
Drop (Clean - Poisoned): 0.0313
Drop (Clean - Init): 0.0830
Drop (Clean DP - Init): 0.0834
Drop (Poisoned - Init): 0.0517
Drop (Poisoned DP - Init): 0.0517
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8821
Clean Recall: 0.8758
Clean F1 Score: 0.8753

Clean DP Precision: 0.8828
Clean DP Recall: 0.8762
Clean DP F1 Score: 0.8757

Poisoned Precision: 0.8528
Poisoned Recall: 0.8445
Poisoned F1 Score: 0.8440

DP Protection Precision: 0.8531
DP Protection Recall: 0.8445
DP Protection F1 Score: 0.8439

Summary

FL Simulation Complete
Task: 57c1ed19-e33d-4620-be6d-bf70b86335d8
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8758
Clean DP Accuracy: 0.8762
Poisoned Accuracy: 0.8445
Data Poison Protection Accuracy: 0.8445
Drop (Clean - Poisoned): 0.0313
Drop (Clean - Init): 0.0830
Drop (Clean DP - Init): 0.0834
Drop (Poisoned - Init): 0.0517
Drop (Poisoned_DP - Init): 0.0517
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8821
Clean Recall: 0.8758
Clean F1 Score: 0.8753
Clean DP Precision: 0.8828
Clean DP Recall: 0.8762
Clean DP F1 Score: 0.8757
Poisoned Precision: 0.8528
Poisoned Recall: 0.8445
Poisoned F1 Score: 0.8440
DP Protection Precision: 0.8531
DP Protection Recall: 0.8445
DP Protection F1 Score: 0.8439